

Q1. Where did you get most stuck? Explain the bug/problem and how you solved it.

During the development of the Asteroids game using PyOpenGL, the most challenging issue I faced was related to user input handling and game interaction logic. Initially, the SPACEBAR was not consistently registering for shooting bullets, and even when bullets were generated, they were not visible due to rendering and color issues. Another major difficulty was implementing collision detection between bullets and asteroids, where asteroids were not disappearing correctly after being hit. I also faced problems with object updates where modifying lists during iteration caused unexpected behavior. I solved these issues by carefully separating update logic into temporary lists, ensuring proper GLUT keyboard callbacks were used, and explicitly setting bullet color to white for visibility. Debugging step-by-step using print statements also helped me identify logical errors.

Q2. What did this project teach you that classroom theory could not? Discuss practical lessons learned.

This project taught me many practical aspects of computer graphics that are difficult to fully understand through classroom theory alone. While theory explains concepts like transformations, rendering, and clipping, implementing them in OpenGL showed me how these concepts interact in real-time applications. I learned how frame updates work in game loops, how user input is processed asynchronously, and how small mistakes in rendering order or coordinate handling can completely break visual output. I also gained experience in debugging graphical applications, managing dynamic object lists, and optimizing performance for smooth animation. Most importantly, I understood how mathematical concepts like vectors and distances directly apply to real game mechanics such as movement and collision detection.

Q3. How does this connect to your career track? Relate your work to jobs, research, or game development.

This project is directly connected to my career interest in computer graphics, game development, and simulation systems. Building an Asteroids clone using PyOpenGL provides foundational experience similar to what is required in entry-level game development roles, where understanding rendering pipelines, physics updates, and real-time input handling is essential. It also connects to research areas such as simulation modeling and interactive visualization systems. The experience of implementing transformations, collision detection, and rendering pipelines helps build skills relevant to graphics programming roles, game engine development, and even fields like virtual reality and scientific visualization. This project strengthens my understanding of how theoretical graphics concepts are applied in industry-level software.

Q4. If you had one more week, what would you add? Describe future improvements.

If I had one more week to improve this project, I would significantly enhance the gameplay experience and realism. I would add asteroid destruction animations such as explosion particles to make collisions more visually engaging. I would also implement rotation-based shooting so that bullets fire in the direction the ship is facing, making the game more realistic like the original Asteroids. Additionally, I would introduce a scoring system with increasing difficulty levels and possibly a life system for the player. Another important improvement would be adding smooth physics-based movement with acceleration and inertia instead of simple grid-based movement. Finally, I would improve the graphics by refining asteroid shapes using better curve generation techniques such as Bezier curves.

Q5. Would you recommend this type of projects in CG lab? Be honest with justifications.

Yes, I would strongly recommend this type of project in the Computer Graphics lab because it provides a perfect balance between theory and practical implementation. Working on a real-time interactive project like a game helps students understand core graphics concepts such as transformations, rendering pipelines, and collision detection much more effectively than traditional assignments. It also improves problem-solving skills and introduces debugging techniques for graphical systems, which are important in both academic and industrial contexts. However, it is also important that such projects are guided properly because beginners may struggle with OpenGL setup and event handling. Overall, it is an excellent hands-on learning experience that makes computer graphics more engaging and meaningful.